START SMARTER

Welcome

WHY GO DEEPER?

Everyone has AI.
Most just press go.
Few understand it.

— *be smarter than the tool.*

Welcome. You're about to learn how the most powerful tool of your lifetime actually works. Not five quick tips. Not yesterday's hype. The real machinery.

WHY WE BUILT THIS

We're Luke and Nate, and yeah, we're still in high school. This started at our kitchen table. AI is in every headline now, and it's going to be in every career, including ours. We're the ones who'll be applying for those jobs, so we'd rather understand it now than scramble to catch up later.

We went looking for something good to learn from, but most of it was shallow, focused on tools, or already out of date. So we built our own. The two of us did the building, while our dad taught us the AI itself and showed us how to actually put a course together.

Building it is what showed us the bigger point. AI is already everywhere in your life, and it'll define the career you haven't started yet.

And here's the part nobody tells you: the better AI gets, the more it pays to be the person who actually understands it. The gap between people who get how it works and people who just type into the box is only going to get wider, and **which side you land on is up to you**. That edge has a name, and it's the most important AI skill there is: **Be Smarter Than the Tool**.

You'll even catch us in a few of the examples later, usually wearing the jerseys of the greatest hockey team in the world.

Here's your path.

1

Work

Learn when and how to use AI.

2

Understand

Open the hood and see how AI does its magic.

3

Avoid

AI isn't perfect, and neither are

you. Things to watch out for.

4

Embrace

AI is changing the world. You'll be ready for it.

5

Build

What you will do to turn AI into your personal edge.

WHAT YOU'LL NEED

The course itself runs in your browser. Nothing to install. The labs use two outside tools, both free: **NotebookLM**, which is free with a Google account, and the free version of **ChatGPT, Claude, or Gemini**.

TRY IT

One last thing before you dive in: six honest questions. Yes or no, no score, no wrong answer. You'll probably answer 'No' to at least a few. That's the point.

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Why Learn AI?

Pretend you're a scribe in the year 1500. All day, you hand-copy the king's proclamations and send them out across the kingdom. Good work. Steady job. Then one morning, you hear that a guy in Mainz built a machine that prints pages a thousand times faster than you can write them. Two choices: pretend it isn't happening, or learn the new machine well enough to run it.

AI is the press. Run it, or someone else will.

AI IS EVERYWHERE

AI isn't something you go visit. It's already in the apps on your phone, the search results you read, and the tools your first job will hand you on day one. It feels like AI went from sci-fi to normal in about two years. **Most of what it does today, it will do better tomorrow.**

AI is already in the apps you use every day

Most of the time, it's making a prediction: what you might tap, say, watch, search for, or need next.

Recommends
Predicts what you might like next.

Spotify, Netflix, TikTok

Navigation
Predicts traffic and arrival time.

Google Maps, Waze

Face recognition
Checks whether a face matches you.

Phone unlock, photo tagging

Voice assistants
Turns your speech into words.

Siri, Alexa, Hey Google

Chatbots
Carries on a conversation with you.

ChatGPT, Claude, Gemini

AND IT'S EARLY

AI is new for everyone. This is YOUR big advantage.

Why you'll thrive in the AI future.

1. **This is YOUR time**

   The adults figuring out AI started about ten minutes before you did. Nobody has a twenty-year head start. With something this big, that almost never happens, and you're showing up right as it lands.

2 **You'll move faster**

Before AI, a career usually looked like this: spend years paying dues, learning the trade, and climbing the ladder. Then, maybe you'd strike out on your own to start a company. AI collapses that runway. What used to take a decade, you can reach for now.

3 **Nothing to unlearn**

The people who got good at the old way have to undo it: break habits, give up the workflow that made them fast, and trust a tool that works nothing like what they're used to. You skip all of that. You're learning the work with AI from the start.

THIS HAS HAPPENED BEFORE

Before personal computers, becoming a designer meant years at a drafting table learning the craft by hand. Then desktop publishing showed up, and a teenager with a Mac could turn out professional work in an afternoon. The tool didn't replace skill, it shortened the distance between wanting to do the work and actually doing it. And it's the rule, not the exception: the steam engine did it for physical labor, electricity for the factory, the internet for information. **AI is that, for almost everything.**

And this isn't just tech industry hype. In July 2025, the White House released an official national strategy document, *Winning the Race: America's AI Action Plan*. Here's how it describes what AI makes possible:

> An industrial revolution, an information revolution, and a renaissance—all at once. This is the potential that AI presents.
>
> — WINNING THE RACE · The White House, July 2025

That's not a forecast for someday. The information revolution is already here. **Today.**

Whatever career you pick, AI will be sitting in the middle of the work. Learn to push AI to its full potential, and you'll run circles around everyone still scratching the surface.

TRY IT

AI can help you reach your goals faster than was ever possible. For each one, decide whether AI shortcuts the work or whether it's still on you.

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What Is AI?

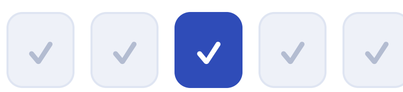
You hear about AI constantly. So what is it, really? The useful starting point: AI is software built to do things that used to take a human brain. And it comes in two kinds.

Types of AI (and you already use both)

SORTS & RECOMMENDS

Recommendation AI

Chooses from what already exists



Ranks the options and picks the top one

WHERE YOU MEET IT

- Next show on Netflix
- Next song on Spotify
- Fastest route on Maps

Never makes anything new.

CREATES NEW CONTENT

Generative AI

Makes something that didn't exist

```
Twin boys in high school, who love the
Dallas Stars, are building an AI course
at a warm kitchen table.
```

This prompt made the picture on the welcome page.

WHAT IT CAN DO

- Writes the email
- Drafts the essay
- Makes the image
- Codes the website
- Creates the song
- Builds the video

When people picture AI, they usually mean this, and it's what this course is about.

Here's what that difference looks like on one task.

THE TASK

"What movie should I watch tonight?"

📺

Recommendation AI

Netflix-style recommendation

~~Midnight Harbor~~ ~~seen~~

Echoes of the Tide TOP PICK

The Glass Orchard

Northern Lights

✨

Generative AI

Writes a reply, in a conversation

How about Midnight Harbor? It's the slow-burn thriller most people rate highest right now, rainy and tense the whole way through.

It just moves to the next title on the list. It has no idea why you skipped the first.

Composes a reply, and you can talk back.

🛋️ I already saw that at a friend's.

Ah, then skip it. Try Paper Cranes instead, it keeps that same rainy, tense mood but it's a quiet little mystery you've almost certainly missed. Want something lighter?

Recommendation AI just slid to the next title on its list. Generative AI took your situation, that you'd already seen it, and composed something new from it. And it's still in the conversation.

THE NAME GAME

You'll hear a pile of names: AI, generative AI, ChatGPT, LLM. Don't let the terminology trip you up. For the rest of this course, we'll usually say AI, and we mean the generative kind that you experience with ChatGPT, Claude, and Gemini.

There's a name for these generative AI tools: LLM, short for **Large Language Model**. Think of it like this. ChatGPT is the app you use, and the LLM is the engine under the hood doing the work.

What's an LLM?

L **Large**

Trained on **huge amounts of text**: books, websites, even code. That scale is what lets one model handle many different tasks.

L **Language**

Built for human language: **reading, writing, summarizing**, translating, explaining. Words in, words out.

M **Model**

The trained AI itself. It takes your input and **predicts a likely output** based on patterns it learned from data.

TRY IT

You probably use both kinds of AI every day. For each example, decide whether it's recommendation AI picking from what already exists, or generative AI making something new.

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How an LLM Works

You’ve met the two kinds of AI and named the generative one: a Large Language Model, or LLM. So how does an LLM actually turn your words into an answer?

When ChatGPT or Claude write a sentence, they're running math to predict the likely next words. They aren't looking up what your words mean; they're working out which words tend to follow which.

How do they do it? In two phases. First the model learns, once, by soaking up patterns from mountains of text. Then, every time you chat, it uses those patterns to build your answer one word at a time.

LEARN ONCE

🌱 **Training**

↓

🍀 **Patterns**

→ patterns power every answer

ANSWER · EVERY WORD

🎲 **Probability**

↓

✍️ **Prediction**

Four ideas carry it. Below, we follow one example, peanut butter, through all four.

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01 🌱 Training

LEARN ONCE

The model **teaches itself**: guess the next word, check, and nudge its numbers toward the right word.

1 · READS

📖 books | 🌐 web | 💬 chats | 💻 code

more than you could read in 1,000 lifetimes

→

2 · GUESSES

`Peanut butter and __`

`cloud` ✗

wrong → nudge the internal numbers

→

3 · CORRECTS

`Peanut butter and __`

`jelly` ✓

a little more accurate every pass

Repeat that loop billions of times, and that's training.

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02 🍀 Patterns

LEARN ONCE

So what is it actually learning? **Patterns**. Here's one you picked up as a child. Which word comes next?

Peanut butter and ——— →

jelly *you knew it, so does AI*

PATTERNS ARE EVERYWHERE

Twinkle, twinkle, little __	→	star
Once upon a __	→	time
Better late than __	→	never

AI didn't memorize these. It absorbed the pattern by running through billions of examples.

---

03 🎲 Probability

EVERY WORD

AI doesn't make one guess. It scores **every** possible next word: a ranked list with a probability on each, and those numbers shift with the surrounding text.

SAME WORD, DIFFERENT ODDS

I'd like to buy peanut butter and _____.

jelly	41%

bread	27%
bananas	16%
honey	5%

I'd like to buy a peanut butter and banana _____.

sandwich	54%
smoothie	16%
toast	9%
jelly	2%

Add the word banana, and jelly's probability drops from 41% to 2%. The word banana changed everything.

---

04 ✍️ Prediction

EVERY WORD

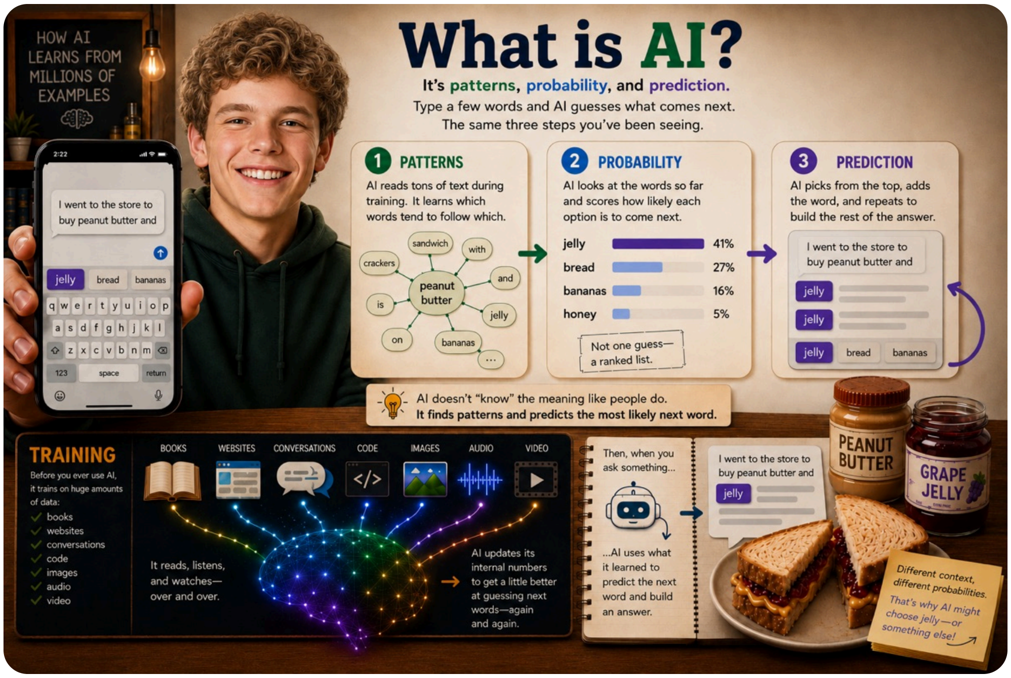
Probability handled one word. But your answer is hundreds of words long, so the model just repeats the move. Your phone does this when you write a text: it suggests a word, you tap it, it suggests the next. AI works the same way, with no fixed plan for where the sentence will end up. One word, look again, the next.

1 I want to buy peanut butter and + jelly

2 I want to buy peanut butter and jelly + for

3 I want to buy peanut butter and jelly for + lunch

A full paragraph runs this loop many times, fast enough to look like thought.



That clears up three big myths:

- AI isn't **magic**: it's math working out probabilities.
- It isn't **a person**: no thoughts, no understanding, even when it sounds like it has both.
- It isn't **a truth machine**: it predicts what sounds likely, so a wrong answer can sound just as confident as a right one.

Keep those three straight and much of the confusion falls away.

TRY IT

You just saw what AI really is: math predicting the next word, and not magic, a person, or a truth machine. For each statement, decide whether it's true or false.

START SMARTER

Does AI Think?

You type a question and AI types back like a person. It explains, it jokes, it says sorry when it slips up. After a while it’s hard not to feel like someone is in there. So here’s the big question: **Does AI actually think?** Is it alive? Is it smart? Does it feel anything?

Short answer: no, not the way you do.

IT’S IN THE NAME

We call it Artificial Intelligence, not Artificial Thinking. That was picked on purpose. Something can act intelligent, finish your sentence, explain a poem, without understanding a single thing it just said.

There’s a famous way to picture this, sometimes called the Chinese Room. Imagine someone locked in a room who doesn’t know a word of Chinese. Notes in Chinese come in under the door. The person can’t read any of it, but they have a giant rulebook telling them which symbols to send back. So they match symbols and slide back an answer that looks perfect to the people outside.

That's AI. It arranges symbols brilliantly and understands none of them. You get a perfect-looking answer with no real understanding behind it.

When you think vs. what AI does

👦 A HUMAN MIND **When you think**		🤖 A PATTERN MACHINE **What AI does instead**
Understand what things actually mean	⟷	Matches patterns in mountains of text
Draw on real experience and senses	⟷	Only ever reads about the world

Choose words to make a point	Predicts the next likely word
Actually care whether you're right	Has no stake in being right or wrong
Feel when something's beautiful	Echoes what people call beautiful
Know when you're unsure	Can't tell when it's making things up

None of this means AI is dumb or useless. A pattern machine this good is genuinely powerful. It just means it's not thinking. Once you see that, its weird moments start to make sense.

🔑 **AI doesn't think. It predicts.** At its core it's a pattern machine on steroids, and the thing running underneath is plain math. That's not an insult to AI. It's the single most useful fact you can carry into the rest of this course.

TRY IT

AI can sound like it thinks, feels, and understands. For each statement, decide whether it's something AI really does, or something that just sounds true.

START SMARTER

What You Can Control

You're hearing it constantly: AI is taking jobs. That headline used to be a prediction. Now it's news.

And jobs are just the loudest headline. AI is also reshaping who holds power, what it costs the planet to run, and what you can trust online. It's all real, it's all moving fast, and almost none of it is in your hands.

WHAT IS GOING TO HAPPEN?

Will it erase millions of jobs or create new ones? Will it wreak havoc on the environment? **Nobody knows yet, including the people making the predictions.**

Right now, there's a big question worth asking: what's in your hands, and what isn't?

Out of your hands		In your hands
Which jobs change first	⟷	What you actually learn
Headlines and hype	⟷	How much the hype gets to you

School and workplace policies	⟷	How hard you work in school
How fast AI improves	⟷	How well you use AI
New laws and regulations	⟷	Whether you keep thinking for yourself

Most of the noise around AI is the left column. Most of the leverage in your life is the right column.

SO WHAT DO YOU DO?

That right column isn't a feeling, it's a to-do list. Three moves worth your energy:

1. 🛠️ **Get genuinely good with AI, not just familiar.**
   Pick one tool and go deep: what it's great at, where it gets things wrong, how to push it. Depth beats dabbling.

2. 🧠 **Do the thinking first, then bring AI in.**
   Form your own take before you ask, then use AI to sharpen it instead of skipping it. That's the line between getting smarter and getting dependent.

3 🔨 **When the hype spikes, close the tab and go build.**
You can't control the headlines. You can control whether you doomscroll them or spend that hour getting better at something real.

Controlling your skill and judgment isn't a consolation prize for the stuff you can't change. It's the one lever that actually tilts your odds, even when the big forces aren't yours.

🔑 **The world will change in ways you don't fully control.** That doesn't make you powerless. You control how this actually goes for you, what you learn, and how well you use the tool. That's where your energy goes.

TRY IT

Some of what AI does to your future is out of your hands. A lot of it isn't. For each one, decide whether it's in your control or not.

START SMARTER

Does School Matter?

A lot of people your age are quietly asking the same thing. If AI can answer anything, do you even need school anymore? Why sit through years of classes when the tool spits out the answer in seconds?

Think about it this way.

Fast forward a few years. You've landed your dream job at Google, and AI sits right next to you all day, cranking out drafts, code, and plans. Your coworker at the next desk is doing the same. You each have the same job, using the same AI.

Here's the catch: ask AI the same question, and it hands back the same answer, to you, your coworker, and everyone else. So what makes you more valuable? It's not the answer itself. **It's what you already know, which shapes the question you ask and what you do with the answer.**

WHAT MOST PEOPLE DO NOT UNDERSTAND

In the AI era, learning is more important than it's ever been. This seems counterintuitive, but it's true. This inspired the name of this course, **Be Smarter Than the Tool.** Whatever college and career you choose, the more you learn, the more successful you'll be.

In almost any job, you'll have AI as your partner. So, for your dream job at Google, here's what it looks like.

Two skills. Both grow with what you know.

AI gives everyone the same answer. These two skills are yours, and the more you learn, the better you get at both.

1

Ask the right question

What you know shapes what you ask. A sharper question gets a better answer, before AI does anything special.

2

Make the answer better

Read it, judge if it's right, push back, and make it better. AI can't do this for you, it doesn't know what you know.

Those two things are your domain. This is where you need to live. And it starts today, not when you work at Google. Double down on building your knowledge and skills. And one day you'll be the CEO at Google.

🔑 **The difference is you.** AI hands similar answers to anyone who asks a similar question. Asking a better question, and making the answer better, only happen if you know enough. That's why you learn.

TRY IT

Here are five answers AI gave, each one stated with total confidence, and each one with something wrong. The question isn't whether AI makes mistakes. It's whether what you've learned lets you catch them. For each, be honest: can you catch the mistake, or would you have to trust it?

START SMARTER

Learn with AI

Here's the good news: when it comes to studying, AI might be the most powerful tool you'll ever get your hands on. Used the right way, it doesn't just help you finish the work, it makes you genuinely sharper at the subject.

It's the most patient tutor you'll ever meet. It'll quiz you at 1am, never sigh when you ask again, and meet you exactly where you're stuck.

The first move is choosing which kind of AI to aim at the job. There's a simple question to guide your choice: do you want to learn from materials you already have, or learn something new?

Which study tool for the job?

📒 NOTEBOOKLM

Source-grounded Tutor

Learns from your materials.

It only knows what you give it, which makes it the ultimate exam-prep machine.

✨ CHATGPT · CLAUDE · GEMINI

General Tutor

Makes things from scratch.

Trained on massive amounts of data, it can explain or quiz you on almost anything.

🎯 **BEST USE** You have the materials the test covers: class notes, screenshots, study guides, even a webpage or YouTube video.	🎯 **BEST USE** You don't have the materials, or you want a concept explained a new way or extra practice your notes don't cover.
⚠️ **THE CATCH** It doesn't know what you didn't give it. If your notes are missing a key concept, it can't reliably fill the gap, because it only works from the materials you gave it.	⚠️ **THE CATCH** It can occasionally make things up, or explain a concept differently than how your teacher taught it and expects on the exam.

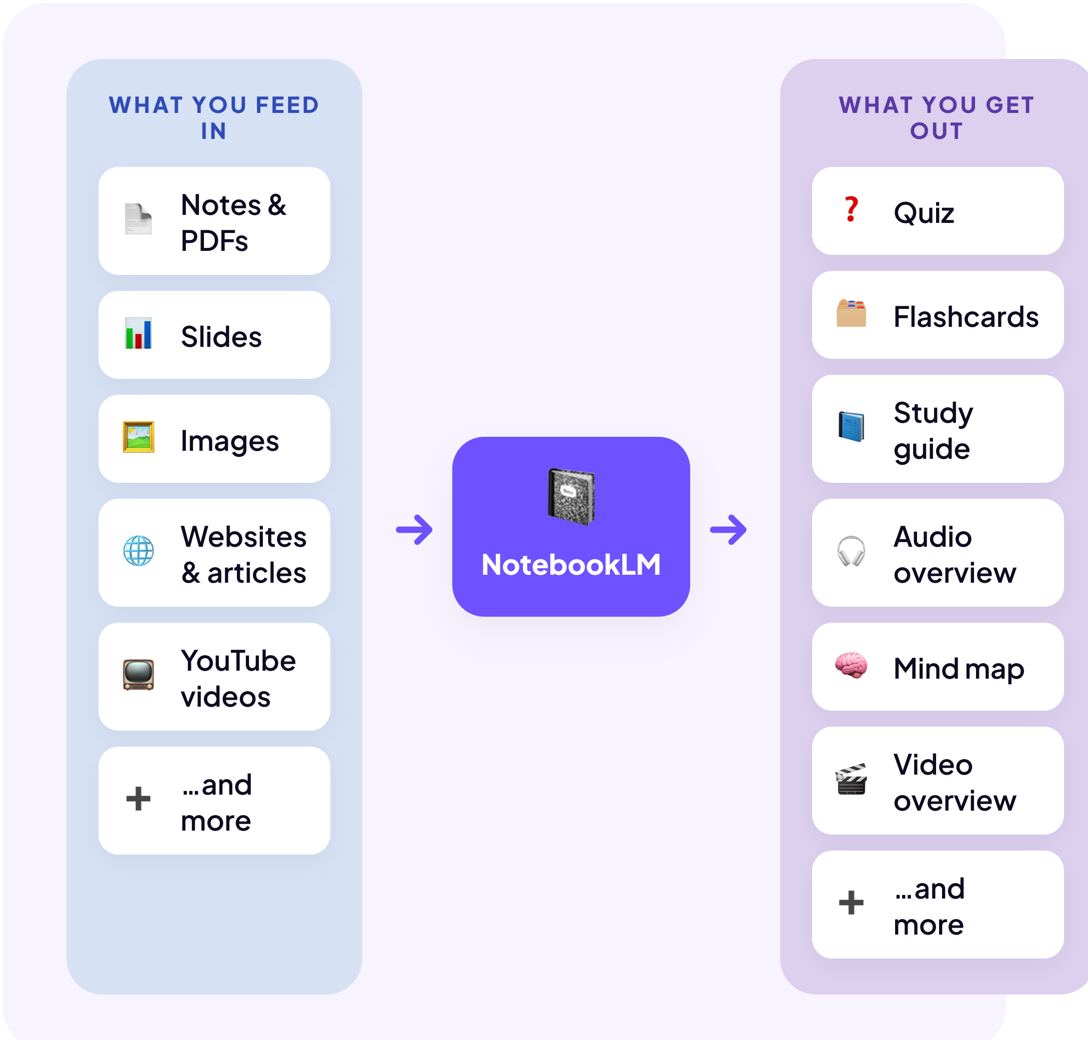
START HERE

Use NotebookLM as your main study tool. The clue is in the name: it was built for learning.

ChatGPT or Claude can do most of the same things, but it's more difficult because they weren't designed with the goal of learning.

HOW NOTEBOOKLM WORKS

Feed it your own study materials, and it turns them into the kind of study tool that fits you best: audio to listen to, quizzes to test yourself, and more. You choose.



BEST PRACTICES

Here's how to use NotebookLM to its full potential.

1 **One notebook per subject**

Keep AP History and Chemistry in separate notebooks. Mixing sources confuses the model and gives you muddled study guides.

Name each one clearly and treat it like its own binder.

2 📚 **Feed it a few angles, not one**

For a big test, give it everything your teacher handed you: your class notes (so it keeps your teacher's wording), the slides, and any video or link the teacher shared. NotebookLM reads a video's transcript and blends it with your notes, so everything it gives back stays true to your class.

3 ❓ **Quiz yourself blind**

Re-reading notes feels productive, but that's recognition, not recall. Generate a quiz, close everything, and take it cold.

```
"Generate a 10-question quiz from these
sources only. Don't show the answers
yet. Wait for mine, then grade me."
```

4 🎧 **Review on the go**

Have NotebookLM make an audio overview, then play it while you drive, walk, or work out. It turns dead time into another pass over the material.

5 🔎 **Trace it back to learn it**

When something doesn't fully click, click its numbered citation to jump straight to the spot in your own notes it came from. Reading the original material is often what makes it land, and it's faster than re-reading the whole chapter.

LAB 01

Your first lab. In about 25 minutes, with a laptop and a free Google account, you'll turn this course into a NotebookLM study notebook and use it to quiz yourself.